



Analysis of shrinkage deformation behavior in microwave vacuum drying of Chinese yam based on evolution of physical properties

Qian Zhou^a, Linlin Li^{a,b,c,*}, Lijuan Wang^d, Wenchao Liu^{a,b,c}, Weiwei Cao^{a,b,c}, Tongxiang Yang^{a,b,c}, Junliang Chen^{a,b,c}, Guangyue Ren^{a,b,c}, Chung Lim Law^e, Xu Duan^{a,b,c,**}

^a College of Food and Bioengineering, Henan University of Science and Technology, 471000, Luoyang, China

^b Henan Province Engineering Research Center of Agricultural Products Processing Equipment, 471000, Luoyang, China

^c Henan Province Engineering Technology Research Center of Agricultural Product Drying Equipment, 471000, Luoyang, China

^d College of Basic Medical Science, Ningxia Medical University, 750004, Yinchuan, China

^e Department of Chemical and Environmental Engineering, Malaysia Campus, University of Nottingham, Semenyih, 43500, Selangor, Malaysia

ARTICLE INFO

Keywords:

Microwave vacuum drying
Shrinkage deformation
Image processing
Physical properties

ABSTRACT

Shrinkage of fruits and vegetables during drying directly affects the sensory quality of products, and understanding its behavior and mechanisms is crucial for quality improvement. This study utilized image processing techniques to analyze the shrinkage behavior of Chinese yam during microwave vacuum drying (MVD). The drying characteristics and physicochemical properties (textural properties, viscoelastic properties, and moisture state) of Chinese yam were investigated under microwave power densities (MPDs) ranging from 2 to 10 W/g to elucidate the mechanisms of shrinkage. Additionally, the color and bioactive compounds (total phenolics and flavonoids) of the dried products were also evaluated. Results showed that the shrinkage behavior during MVD exhibited a two-phase pattern: a rapid shrinkage phase followed by a slower shrinkage phase, with a distinct inflection point (moisture content range of 0.82–1.19 g/g dry basis) dependent on moisture dynamics. Volumetric shrinkage resulted from combined radial and longitudinal contraction. Significant longitudinal shrinkage manifested as central collapse was observed at low MPDs (2 and 4 W/g). The radial shrinkage (top-view projected area) ranged from 39.76 % to 41.81 % across conditions, showing no significant differences. The rapid loss of free water was the primary driver of shrinkage, while the increase in sample hardness (textural property) and residual modulus (viscoelastic property) reflected the enhancement of mechanical strength, effectively resisting shrinkage induced by drying. Additionally, high MPDs not only improved volume retention but also demonstrated favorable effects on color preservation and retention of bioactive compounds.

1. Introduction

Chinese yam (*Dioscorea opposita* Thunb), a tuber with dual medicinal and edible value, is known to enhance immunity and regulate metabolism (Li et al., 2022a, 2023). However, its high moisture content (MC) makes it prone to spoilage during storage. Drying is necessary to extend shelf life and reduce transportation costs (Li et al., 2024; Liu et al., 2022; Ren et al., 2022). Conventional hot-air drying causes severe shrinkage and low efficiency, while freeze-drying's high cost and prolonged duration hinder practical application (Duan et al., 2016; Zhang et al., 2023). Microwave vacuum drying (MVD) combines vacuum drying's

low-temperature evaporation with microwave rapid heating, effectively preserving quality while improving drying efficiency (Ando and Nei, 2024; Li et al., 2019). This energy-saving method has been successfully applied to sea cucumbers, pumpkins, and apples (Ando et al., 2019; He et al., 2021; Monteiro et al., 2018), demonstrating superior performance in maintaining texture, nutrient retention, and rehydration properties compared to traditional methods. However, volumetric shrinkage of MVD products is still evident, leading to reduced rehydration performance and poor texture, thus seriously affecting product quality and consumer acceptance (Seerangurayar et al., 2019; Yang et al., 2024; Zhang et al., 2017).

* Corresponding author. College of Food and Bioengineering, Henan University of Science and Technology, 471000, Luoyang, China.

** Corresponding author. College of Food and Bioengineering, Henan University of Science and Technology, 471000 Luoyang, China.

E-mail addresses: linlinli2020@126.com (L. Li), duanxu_dx@163.com (X. Duan).

Existing studies on the shrinkage behavior of fruit and vegetables during drying have obvious limitations, with most studies focusing on final shrinkage measurements or numerical simulations. For example, Ortiz-García-Carrasco et al. (2015) examined potato shrinkage using image processing, while Chao et al. (2023) reported less deformation in far-infrared drying after 87.29 % free water removal, but neither elucidated the full shrinkage mechanism. Yuan et al. (2019) identified peak shrinkage during intermediate drying stages, attributing it to moisture stress. Current research lacks systematic exploration of shrinkage evolution and true mechanisms, and there is an urgent need to combine the drying stage and conditions to analyze the intrinsic correlation.

Shrinkage of food during drying is caused by a number of factors. Cellular water migration leads to pore collapse and tissue deformation, while different states of moisture removal critically affect the degree of shrinkage, but the intrinsic mechanisms remain unclear (Mahiuddin et al., 2018). In addition, dynamic changes in moisture and temperature fluctuations alter the viscoelastic and textural properties of materials (Gulati and Datta, 2015; Mahiuddin et al., 2018), which determine the shrinkage morphology and volume of the final product. Based on this, the present study innovatively used image processing techniques to achieve dynamic resolution of the multidimensional shrinkage behavior of yam during MVD. Through the systematic analysis of drying characteristics, physical properties and moisture state, the intrinsic mechanisms affecting shrinkage were investigated in depth. Additionally, the changes in color and bioactive compounds of the dried products were evaluated, which provides a theoretical basis for optimizing the MVD process.

2. Materials and methods

2.1. Materials

Fresh Chinese yam was purchased from Wen County, Jiaozuo City, Henan Province. Samples that are intact without mechanical damage were selected as experimental material. The initial moisture content, measured by oven drying at 105 °C, ranged from 2.57 to 3.16 g/g (dry basis, d.b.). Before the drying experiments, the yams were peeled, washed, and wiped of surface moisture. Then they were sliced into cylindrical shapes measuring 20 ± 0.5 mm in diameter and 5 ± 0.2 mm in thickness using a mould.

2.2. Drying procedure

Drying experiments were conducted using an intelligent microwave vacuum system (Jinbaili Microwave Equipment Co., Ltd., Nanjing, China). Five microwave power densities (MPDs) -2, 4, 6, 8, and 10 W/g-were applied under a constant vacuum pressure of -0.08 MPa. The system was equipped with an automatic temperature control function. During drying, the chamber temperature increased rapidly after microwave activation. When the temperature exceeded 55 °C—typically in the final drying stage with low moisture content—the system briefly shut off the microwave power to maintain the sample temperature around 55 °C. The preset power level was then promptly restored.

In the experimental operation, 50 g of yam slices were arranged in a single layer on a glass tray and subjected to MVD under predetermined parameters (microwave power, maximum temperature, and vacuum pressure). During drying, samples were periodically removed for rapid weighing and data recording until the moisture content reached below 0.1 g/g (d.b.), at which point drying was terminated.

2.3. Determination of moisture content

Samples were dried by direct drying method at 105 °C in an electrically heated constant temperature blast drying oven until reaching constant weight. The test was repeated three times.

2.4. Drying characteristics

The drying rate (DR) of the sample during the drying process was calculated using equation (1):

$$DR = \frac{M_t - M_{t+\Delta t}}{\Delta t} \quad (1)$$

where M_t is the moisture content at any drying time t , $M_{t+\Delta t}$ is the moisture content at time $t + \Delta t$, and Δt is the time difference between measurements.

2.5. Determination of volumetric shrinkage

The sample volume was measured using the quartz sand displacement method (Xu et al., 2022a). The tests were performed three times and the volume shrinkage ratio (VSR) was determined using the following formula:

$$VSR(\%) = \frac{V_0 - V_t}{V_0} \times 100 \quad (2)$$

where V_0 is the initial volume of the fresh sample and V_t is the volume at time t during the drying process.

2.6. Shrinkage deformation analysis based on digital image techniques

Top-view projected area shrinkage ratio (TPASR) was measured following the method of Yang et al. (2024). Images of the top-view projected surface of the samples were taken using a digital camera (IMX 800, Sony Corporation, Tokyo, Japan). The images were then binarized using MATLAB (Matlab 2024, MathWorks Inc., Natick, MA, USA) to obtain sample edges. The edges were subsequently extracted in a closed-fill operation on the edges and the area enclosed by the contour curves was calculated. Each test was repeated three times. The following formula was used for the calculation of TPASR:

$$TPASR(\%) = \frac{S_0 - S_t}{S_0} \times 100 \quad (3)$$

where S_0 is the initial projected area of the sample and S_t is the projected area at any drying time t .

2.7. Determination of textural properties

Following the method of Li et al. (2024), textural properties including hardness and crispness were measured using a TA-XT2i texture analyzer (S4800, Stable Micro Systems Ltd., UK) under different drying conditions. A cylindrical probe P2 with a diameter of 2 mm was used for the testing procedure. During the test, set the test speed to 1 mm/s, the trigger force to 0.05 N and the penetration depth to 1 mm to record the force and distance curves. Hardness and crispness are determined by the maximum force and the number of force peaks and the slope of the first peak, respectively. Six replicates were performed for each sample.

2.8. Stress relaxation test

The stress relaxation behavior of the samples was tested using a texture analyzer according to the method of Li et al. (2022b). The stress relaxation behavior of yams at different stages of drying were recorded by compression at a constant strain of 15 % and a speed of 1 mm/s for 120 s during the test. Each test was repeated three times. The modified Maxwell model (equation (4)) was applied for curve fitting:

$$\sigma = E_0 \varepsilon_0 + E_1 \varepsilon_0 e^{-t/\tau} \quad (4)$$

where σ is the pressure on a sample at time t . E_0 , E_1 present the residual

modulus. and relaxation modulus, respectively. ϵ_0 is the strain compression. τ is the relaxation time. n is the nonlinear index between 0 and 1.

2.9. Moisture state analysis

The moisture state of the samples was analyzed using a modified version of the method outlined by [Chao et al. \(2023\)](#). A low-field nuclear magnetic resonance (LF-NMR) analyzer (MINI20-015 V-I, Shanghai Niumag Electronic Technology Co., Shanghai, China) was employed for the analysis. The Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence was used to collect the decay signals. The key sampling parameters were set as follows: sampling frequency = 200 kHz, main frequency = 21 MHz, sampling points = 900052, waiting time = 3000 ms, 180° pulse time = 28 μ s, number of echoes = 9000, echo time = 0.5 ms, and number of scans = 8. The CPMG decay curves were processed using multi-exponential fitting analysis with MultiExp Inv software (Niumag Electric Co., Shanghai, China). Each group was measured in triplicate.

2.10. Color measurement

The color parameters of dried Chinese yam were measured using a colorimeter (LS170, Shenzhen Linshang Technology Co., Ltd., China) according to the method described by [Xu et al., 2022a](#). The L^* , a^* and b^* represent blackness (0)/brightness (100), greenness (−)/redness (+) and blueness (−)/yellowness (+), respectively. Six replicates of measurements were performed.

2.11. Determination of total phenolic and total flavonoid contents

The extraction of total phenolics and flavonoids from Chinese yam was performed according to the method described by [Li et al. \(2021\)](#). Briefly, 0.5 g of finely ground yam powder was accurately weighed and transferred into a 50 mL centrifuge tube. A volume of 25 mL of 80 % (v/v) methanol was added. The sample was subjected to ultrasonic-assisted extraction at room temperature for two cycles of 15 min each. After sonication, the mixture was placed on a shaker at 150 rpm and shaken overnight at room temperature. The extract was then centrifuged at 10,000 rpm for 10 min. The resulting supernatant was collected and stored in the dark at 4 °C for subsequent analysis of total phenolic and flavonoid contents.

The total phenolic content (TPC) was determined based on the Folin–Ciocalteu method with slight modifications ([Li et al., 2021](#)). Briefly, 0.5 mL of the sample extract was transferred into a centrifuge tube, followed by the addition of 0.5 mL of Folin–Ciocalteu reagent. After reacting for 3 min, 1.5 mL of 7.5 % (w/v) Na_2CO_3 was added, and the mixture was diluted to a final volume of 5 mL with distilled water. The mixture was thoroughly mixed and then allowed to stand in the dark for 60 min. The absorbance was measured at 760 nm using a spectrophotometer. TPC was quantified using a calibration curve prepared with gallic acid at various concentrations, and the results were expressed as milligrams of gallic acid equivalents (GAE) per 100 g of dry weight (mg GAE/100 g DW).

The total flavonoid content (TFC) was determined according to the method of [Liu et al. \(2021\)](#) with slight modifications. 1 mL of the sample extract was mixed with 0.4 mL of 5 % NaNO_2 . After a 5 min reaction, 0.4 mL of 10 % $\text{Al}(\text{NO}_3)_3$ was added, followed by the addition of 3.2 mL of 4 % NaOH. The mixture was thoroughly mixed and incubated in the dark for 60 min. The absorbance was measured at 510 nm. TFC was quantified using a rutin standard curve, and the results were expressed as milligrams of rutin equivalents (RE) per 100 g of dry weight (mg RE/100 g DW). All experiments were performed in triplicate.

2.12. Statistical analysis

All tests were conducted in triplicate. The experimental data were

presented as mean $\pm$ standard deviation (SD). One-way analysis of variance (ANOVA) and Duncan's multiple range test were conducted using SPSS 26 (SPSS, IBM Inc., Chicago, IL, USA) to determine the statistically significant differences of the experimental data at 95 % confidence level ($P < 0.05$). Origin 2022 (Origin, OriginLab Corporation, Northampton, MA, USA) was used for data processing.

3. Results and discussion

3.1. Drying characteristics

[Fig. 1](#) illustrates the drying characteristics of Chinese yam under different MPDs during MVD. Under all MPD conditions, the MC of the samples reduced progressively as drying time increased. Higher MPD resulted in greater drying rates. This phenomenon was primarily attributed to the enhanced absorption of microwave energy by water molecules at higher microwave power levels, thus increasing the impetus for moisture transfer and consequently speeding up the drying process ([Li et al., 2019](#)). As the MPD increased, the time needed to dry

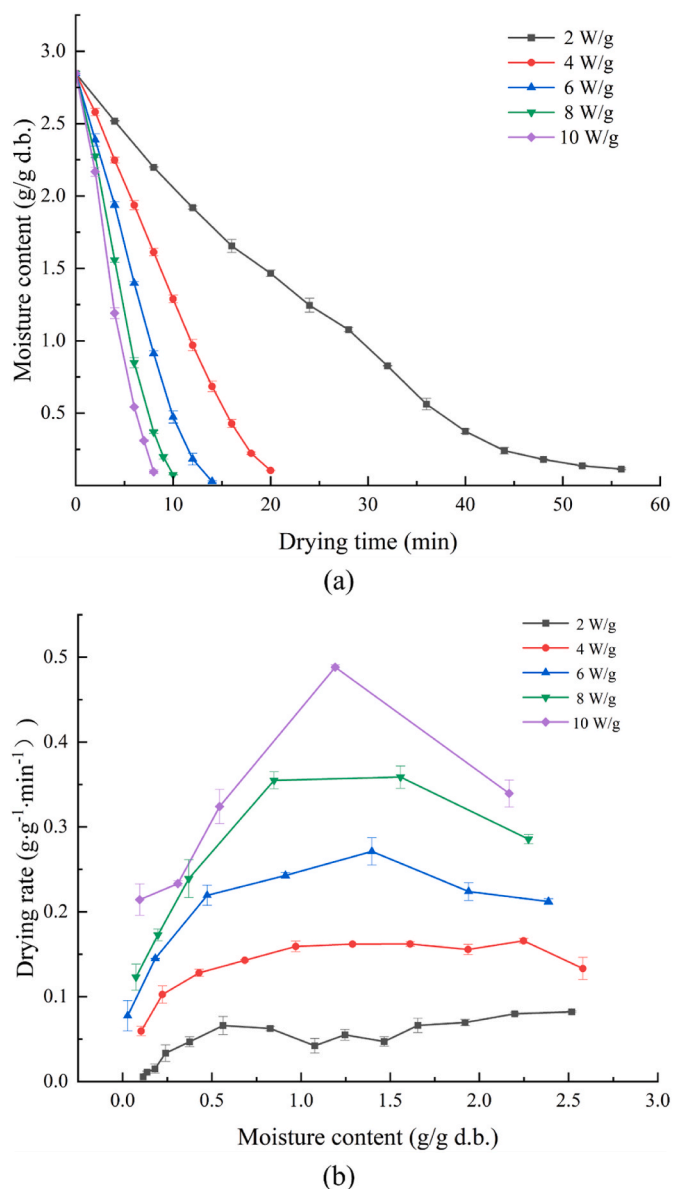


Fig. 1. Drying characteristics of Chinese yam by MVD: (a) drying curves, (b) drying rate curves.

Chinese yam to a safe MC was significantly shortened. Specifically, drying time at 10 W/g was shortened by 85 % in comparison with the time at 2 W/g. The main reason for this was that higher MPD intensified the friction and vibration of the water molecules, accelerating the evaporation and spread of water in the food material, resulting in a significant reduction in the drying time (Li et al., 2018). As shown in Fig. 1(b), a brief accelerating drying phase was observed at the beginning of the drying process when MPD exceeded 2 W/g, which was probably caused by the swift evaporation of surplus moisture from the

surface of the sample (Liu et al., 2022). However, as drying progressed, the microwave absorption capacity of Chinese yam diminished due to the reduction in MC, resulting in a slower drying rate. This falling-rate drying process became more evident at higher MPD levels. As the gradual disappearance of free water on the sample surface occurred, it led to the formation of a relatively compact porous network structure, which impeded the transfer rate of water vapor inside the sample (Sun et al., 2021). Moreover, shrinkage deformation during drying increased diffusion resistance for internal moisture transport, while bound water

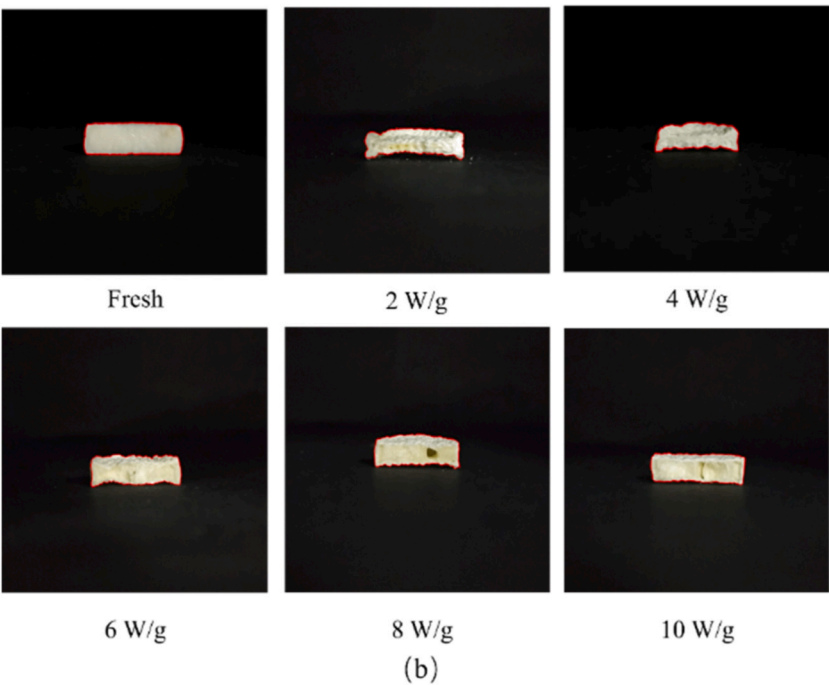
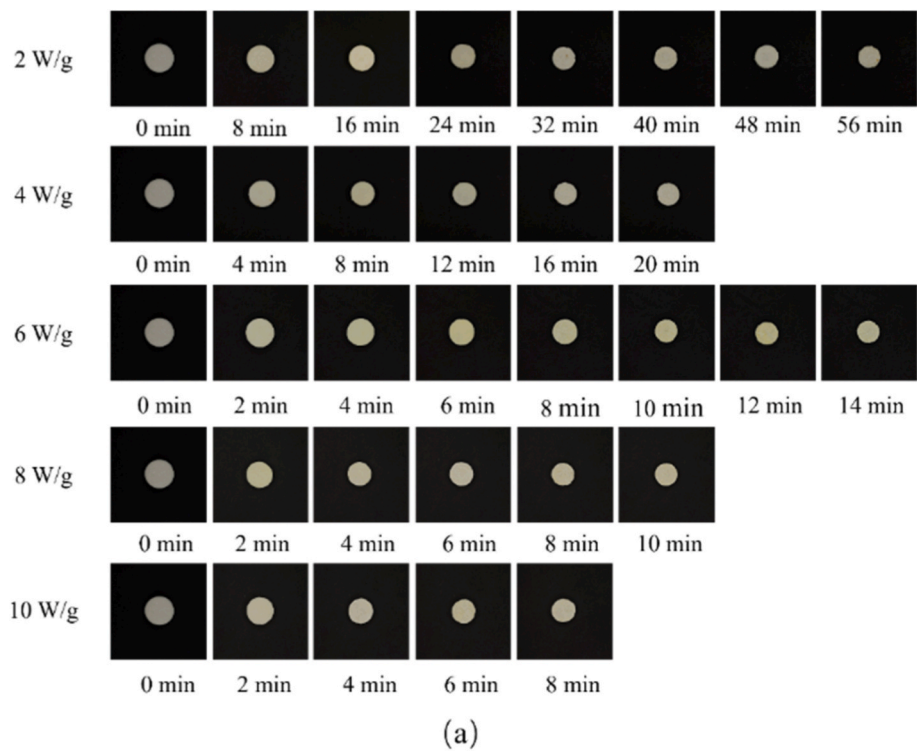


Fig. 2. The morphological characteristics of yam samples under different drying conditions: (a) The top-view projection images of the samples at different times during MVD, (b) Longitudinal section images of fresh and dried samples subjected to different microwave power densities.

in later stages became increasingly difficult to remove, both of which collectively contributed to the continuous decline in drying rate.

3.2. Analysis of shrinkage deformation behavior

Shrinkage is a typical occurrence in dried products and is often considered a significant factor for evaluating the final product's appearance (Chao et al., 2023). This study seeks to investigate the shrinkage deformation behavior of Chinese yam during the MVD process by analyzing both two-dimensional and three-dimensional shrinkage changes.

3.2.1. Analysis of two-dimensional shrinkage behavior based on image processing

Fig. 2 illustrates the morphological characteristics of Chinese yam during the MVD process under various drying conditions. It is evident that the radial and longitudinal shrinkage deformation of the samples varied under different MPD drying conditions. From the top-view projection images of the samples during the MVD process (Fig. 2(a)), it was noted that as the drying continued, the size of the samples under different MPD conditions decreased to varying degrees. This was caused by the shrinkage stress generated by the imbalanced pressures between

internal and external samples during drying, which led to the inward shrinkage of the solid matrix (Aprajeeta et al., 2015; Yang et al., 2024).

To more clearly illustrate the differences in the shrinkage behavior of samples under various drying conditions, image processing was used to extract the contour curves of the top-view projection images of the samples at various drying stages, as shown in Fig. 3. As the MPD increased, the shrinkage rate of the samples accelerated, manifested by significant changes in the contour curves within a shorter time. Conversely, at lower MPD levels, the changes in the contour curves of the samples were more gradual. This difference highlighted the impact of MPD on the rate of moisture and heat transfer inside the samples, thereby affecting their shrinkage behavior. As drying progressed, both moisture and heat were transferred from the interior to the exterior of the samples. Since the moisture in the outer layer was lost first, which also carried away heat, a gradient distribution of moisture and temperature formed within the samples. This gradient distribution generated internal stress, leading to samples to shrink gradually toward the center. As a result, the area and circularity of the samples gradually decreased. In the final stages of drying, the changes in the contour curves of the samples became less pronounced, which may be attributed to surface hardening and the shift of the material state from rubbery to glassy, which inhibited further shrinkage (Gulati and Datta, 2015).

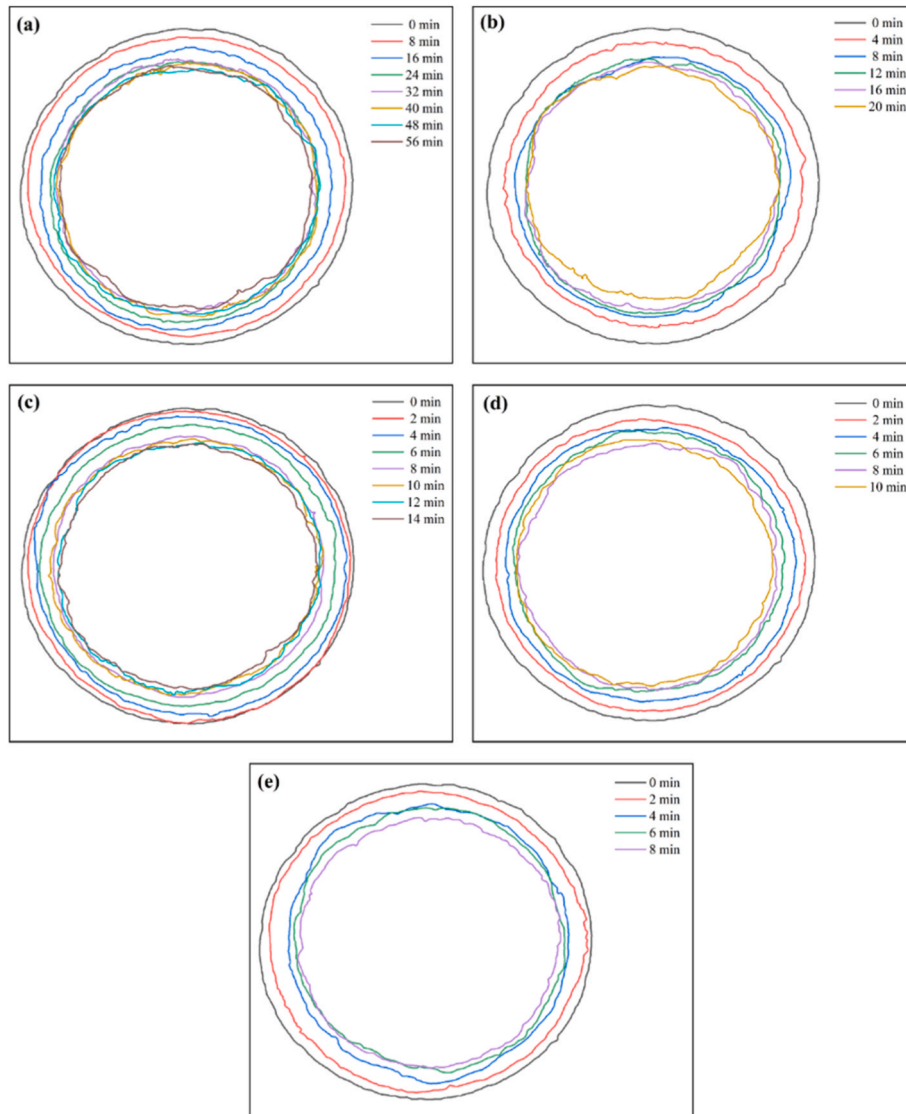


Fig. 3. Contour curve of the projected area of the sample under different microwave power densities of MVD: (a) 2 W/g, (b) 4 W/g, (c) 6 W/g, (d) 8 W/g, (e) 10 W/g.

Fig. 4(a) illustrates the changes in the top-view projected area of samples during the MVD process under different MPDs. For the sample with an MPD of 2 W/g, TPASR initially increased rapidly and then slowed down as the drying process progressed. Samples at other power levels (4, 6, 8, and 10 W/g) also exhibited similar characteristics with rapid shrinkage in the early drying stages followed by a gradual reduction in shrinkage changes. At the end of drying, TPASR of the samples ranged from 39.756 % to 41.81 %, with no significant differences in shrinkage rates observed across different MPD conditions ($P > 0.05$). This suggests that, despite variations in MPD, the impact on the final TPASR of the samples was minimal.

Fig. 2(b) presents the longitudinal section images of dried samples under different microwave power levels. The shrinkage of Chinese yam was primarily characterized by the inward collapse and bending of the middle section of the contour edges toward the center of the sample, as well as the curling of the edges. Similar phenomena have been observed in studies on potatoes (Ortiz-García-Carrasco et al., 2015). Notably, significant central collapse was observed on the sample surfaces dried at MPD levels of 2 and 4 W/g. However, when the MPD exceeded 4 W/g, no collapse or significant changes in height were observed on the sample surfaces. This indicated that drying-induced shrinkage resulted from both radial and longitudinal contraction. At higher microwave power levels, the rapid drying rate and shorter drying time led the solid matrix to transition quickly from an elastic state to a glassy state, thereby reducing the extent of collapse and curling (Mayor and Sereno, 2004). This rapid transition might inhibit structural collapse during the drying process. Additionally, studies have shown that microwave expansion is a primary force in enhancing volume retention. During microwave drying, the intense microwave energy caused partial vaporization of internal moisture, and the swelling pressure of the water vapor partially

counteracted the external pressure-induced shrinkage, thereby reducing structural collapse (Sun et al., 2022; Zielinska et al., 2018). Different drying processes had distinct impacts on the longitudinal and radial shrinkage of fruits and vegetables. During testing, slight surface cracking was observed in samples dried at a power of 10 W/g. Therefore, an MPD of 8 W/g was beneficial for improving volume retention and inhibiting quality deterioration in dried fruit and vegetable products.

3.2.2. Three-dimensional volume shrinkage behavior analysis

The variation in three-dimensional volumetric shrinkage of different MPD-dried Chinese yam is shown in Fig. 4(b). The experimental results revealed that samples at all power levels exhibited significant volumetric shrinkage in the early drying stage, followed by a gradual decrease in shrinkage rate. Notably, samples treated at 2 and 4 W/g demonstrated uniform shrinkage characteristics in the later drying stage. Further analysis indicated that material shrinkage primarily took place during the early and middle stages of drying when the MC remained high. Sun et al. (2022) observed comparable results, showing that the volume retention rate of carrots varied significantly at high MC. During drying period with high MC, liquid moisture moved from the core to the surface, forming capillary menisci. The pressure difference across these menisci generated substantial contraction forces, leading to pronounced shrinkage (Sun et al., 2022). As the MC decreased, the reduction in MC caused a gradual increase in the glass transition temperature of the material. This phenomenon triggered a phase transition from a rubbery state to a glassy state, which enhanced material's rigidity and resistance to shrinkage, consequently slowing or even halting the shrinkage rate (Kurozawa et al., 2012; Sappati et al., 2017). From this, it can be seen that the dynamic influence of moisture migration on material structure, highlighting the complex mechanisms underlying

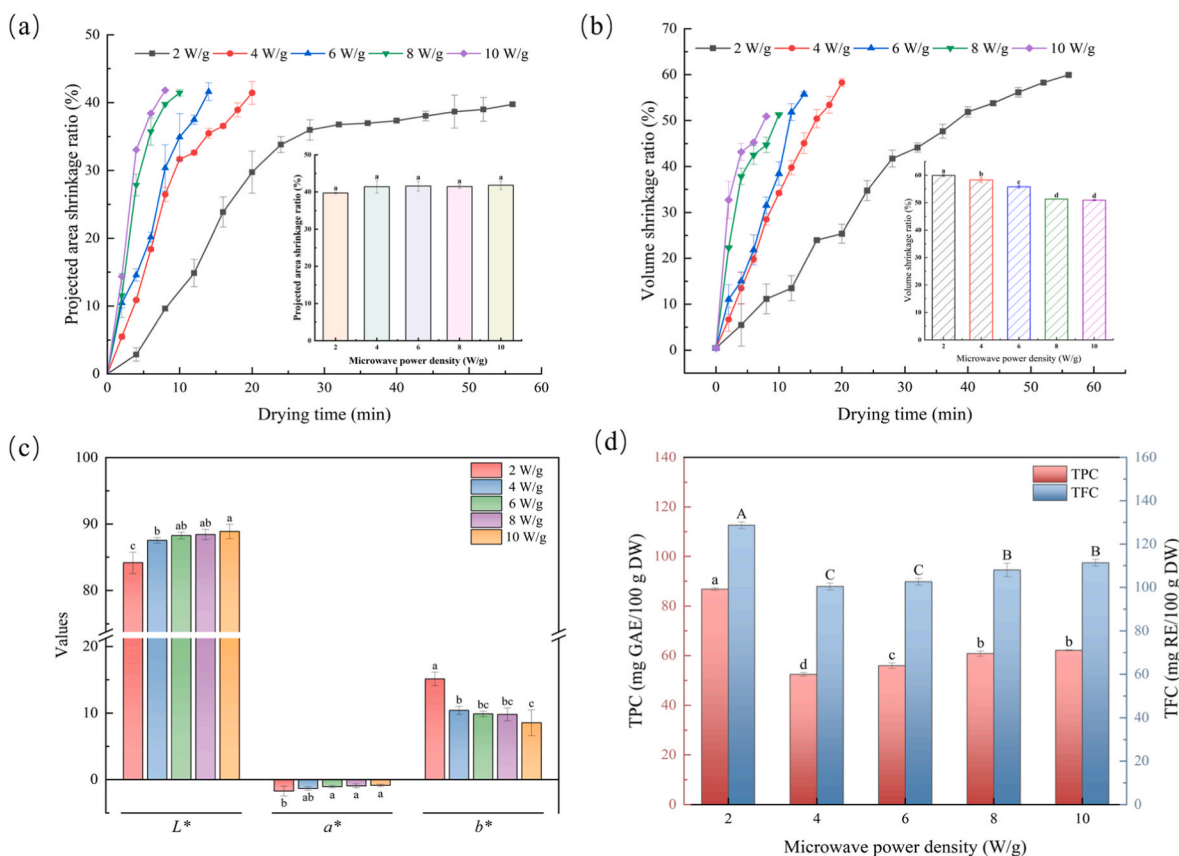


Fig. 4. The shrinkage characteristics and physicochemical properties of Chinese yam at different microwave power densities: (a) top-view projected area shrinkage ratio, (b) volume shrinkage ratio, (c) color parameters, (d) the content of total phenolic and total flavonoid. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

shrinkage behavior during drying.

The final VSR of different MPD-dried products ranged from 50.92 % to 59.931 %, with the sample dried at 10 W/g exhibiting the least shrinkage and the sample dried at 2 W/g showing the most significant shrinkage ($P < 0.05$). This might be due to the longer drying time at low MPD levels. During the extended drying process, the cell turgor pressure gradually decreased, while the accumulation of internal thermal stress led to structural damage to the cell walls or membranes, ultimately

resulting in cell collapse (Mahiuddin et al., 2018). The cell wall, as the primary supporting structure of fruit and vegetable tissues, experienced degradation of its components (such as pectin, cellulose, and hemicellulose) under prolonged thermal stress, resulting in a significant decline in the mechanical strength of the cell wall (Wang et al., 2020), thereby exacerbating shrinkage. With the increase in MPD, the temperature of all samples rose rapidly during the initial drying stage, accelerating moisture diffusion and rapid evaporation. This resulted in a significant

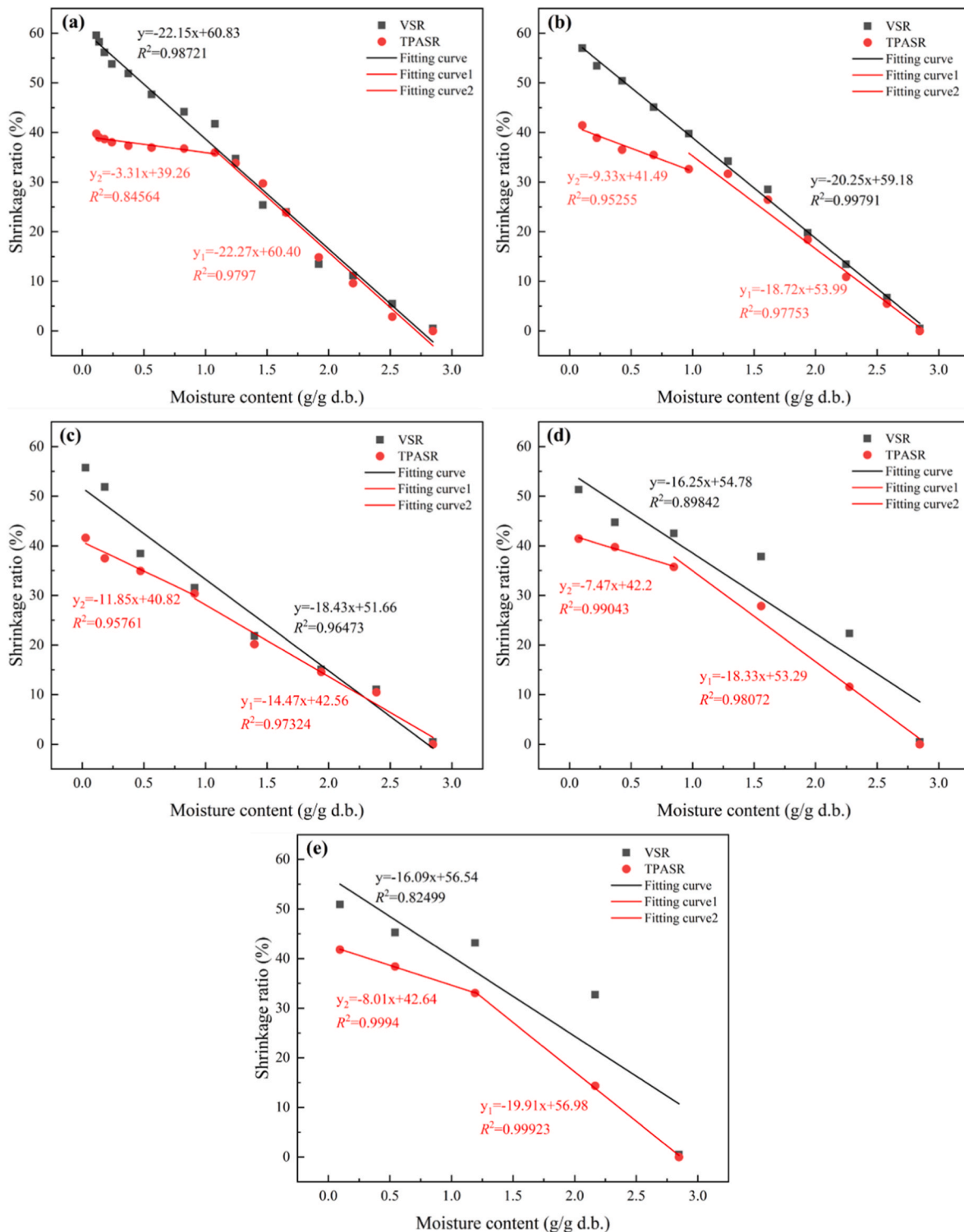


Fig. 5. Linear regression model of moisture content and VSR and TPASR: (a) 2 W/g, (b) 4 W/g, (c) 6 W/g, (d) 8 W/g, (e) 10 W/g (VSR and TPASR indicated volume shrinkage ratio and top-view projected area shrinkage ratio, respectively).

increase in the pressure difference between the inside and outside of the material in a short period of time, which caused a significant shrinkage of the material (Vallespir et al., 2019). As the drying process progressed, as illustrated in Fig. 2(b), high MPD caused water to quickly vaporize into steam. The rapid volume expansion of steam, occurring at a rate exceeding its diffusion speed, enlarged the spatial structure of the material (Huang et al., 2021). This phenomenon inhibited the longitudinal shrinkage deformation of the samples and, to some extent, improved the volume retention rate of the final products.

3.2.3. Analysis of the relationship between shrinkage behavior and moisture content

Based on the experimental results above, linear regression models were established to correlate MC with both VSR and TPASR during MPD

drying, as illustrated in Fig. 5. The findings revealed that as drying progressed, the MC gradually decreased, and the top-view projection area of the samples showed a pattern of rapid shrinkage followed by slower changes. A distinct shrinkage inflection point was observed at a MC range of 0.82–1.19 g/g d.b. Sun et al. (2022) also found moisture-dependent shrinkage inflection points in their studies on carrots, bananas, and king oyster mushrooms. The TPASR exhibited a strong linear correlation with moisture content on both sides of the inflection point, with determination coefficients (R^2) ranging from 0.8456 to 0.9994. Correlation analysis between three-dimensional VSR and moisture content demonstrated that for low MPD levels of 2, 4, and 6 W/g, the volumetric shrinkage of Chinese yam was highly correlated with moisture content, as evidenced by well-fitted regression models ($R^2 > 0.96$). However, at higher MPD levels of 8 and 10 W/g, the

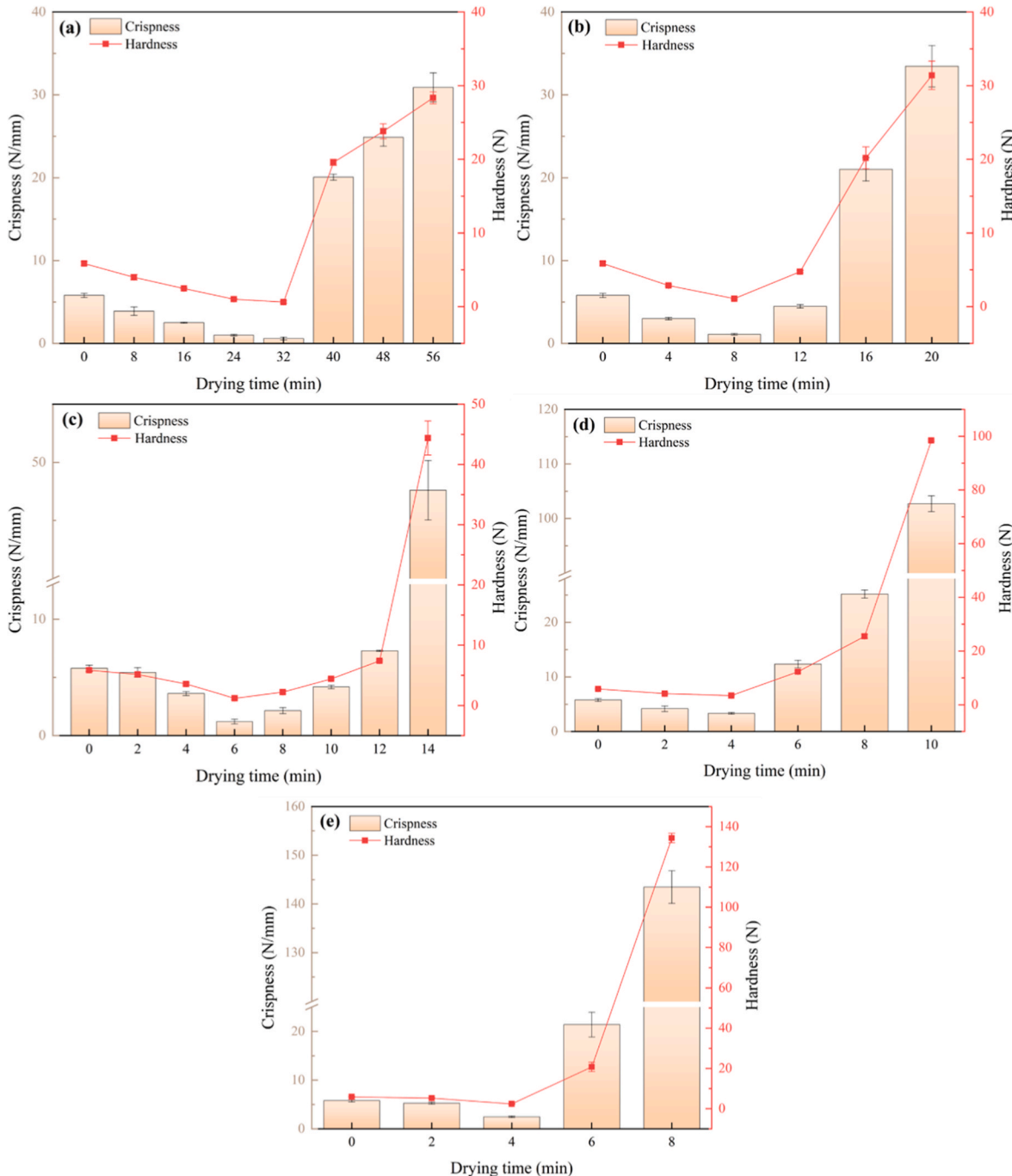


Fig. 6. Variations in textural properties of Chinese yam during MVD: (a) 2 W/g, (b) 4 W/g, (c) 6 W/g, (d) 8 W/g, (e) 10 W/g.

dependence of volumetric shrinkage on moisture content deviated markedly from linear behavior. This deviation may result from the rapid collapse of cellular structure under high MPD. These fitting results suggested that dimensional changes in the product were influenced not only by the amount of evaporated water but also by the structural integrity and collapse resistance of the cellular framework (Ortiz-García-Carrasco et al., 2015). The rigidity of cell walls enabled samples to resist shrinkage, thus reducing macroscopic volume shrinkage (Yang et al., 2024).

3.3. Mechanism analysis

3.3.1. Textural properties

Brittleness, a critical textural attribute of food, reflects a sample's resistance to fracture under bending stress, while hardness characterizes its ability to withstand external forces penetrating its surface. These properties collectively form the mechanical foundation for materials to resist shrinkage deformation, directly influencing their deformation behavior during drying. This study explored the connection between shrinkage behavior and textural characteristics in dried food products

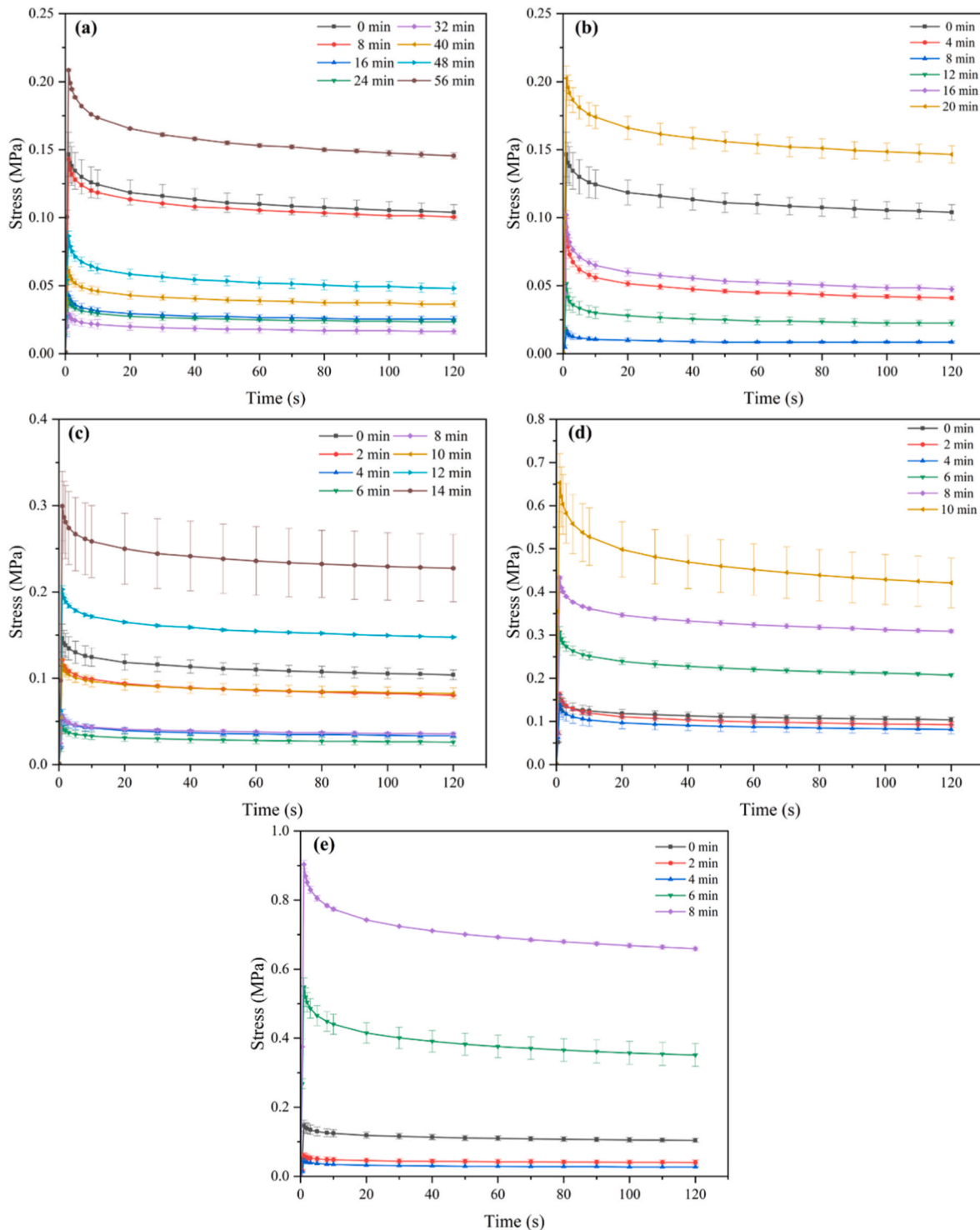


Fig. 7. Stress relaxation curves of Chinese yam during MVD: (a) 2 W/g, (b) 4 W/g, (c) 6 W/g, (d) 8 W/g, (e) 10 W/g.

subjected to MVD.

Fig. 6 illustrates the variations in hardness and brittleness of samples under different MPDs during the MVD process. Both hardness and brittleness under different MPDs exhibited a trend of initial decrease followed by an increase during drying. In fresh samples, the intact cellular structure and sufficient water within the tissue endowed the sample with high mechanical strength, resulting in higher hardness and brittleness. In the early drying stage, moisture loss led to a reduction in intracellular pressure, while microwave heating caused a rise in sample temperature, inducing thermal softening of cell wall components (such as pectin and cellulose). This further exacerbated tissue softening and a decline in mechanical properties, leading to a significant reduction in hardness and brittleness (Duan et al., 2020; Yang et al., 2025). At this stage, the material's inherent mechanical strength was insufficient to withstand drying stresses, resulting in substantial deformation. As drying progressed and moisture content further decreased, the porous structure gradually formed and expanded within the samples. This porous structure enhanced the material's internal support, providing additional rigidity. Simultaneously, the significant reduction in moisture content restricted the motility of molecular chains and elevated the glass transition temperature of the samples, thereby triggering a phase transition from a rubbery to a glassy state. This phase transition significantly enhanced material's rigidity (Ishibashi et al., 2022; Kurozawa et al., 2012; Nguyen et al., 2018). Consequently, the hardness and brittleness of the samples gradually increased. The enhanced hardness and brittleness improved the material's ability to resist drying stresses, thereby effectively inhibiting shrinkage deformation.

Notably, under the MPD of 10 W/g, the sample exhibited the lowest hardness (2.38 N) and brittleness (2.46 N/mm) at 4 min of drying. Compared to the hardness and brittleness of the samples dried at 2 W/g for 32 min (0.62 N and 0.59 N/mm, respectively), these values increased by 2.83-fold and 3.16-fold, respectively. Subsequently, both hardness

and brittleness of the samples showed an upward trend as drying progressed. The rapid increase in hardness and brittleness of the 10 W/g samples may be attributed to rapid dehydration induced by high microwave power, which led to the creation of a compact layer on the surface of the material. Simultaneously, the accumulation of internal steam pressure promoted the development of a porous structure, thereby significantly enhancing material's rigidity. The changes in mechanical strength influenced the shrinkage deformation behavior of the samples, improving their resistance to deformation during the later drying stages.

3.3.2. Stress relaxation behavior analysis

In this study, the changes in the viscoelastic mechanical properties of Chinese yam during MVD were analyzed through stress relaxation tests. Fig. 7 presents the stress relaxation curves of yams under different MPD conditions during MVD. During stress relaxation measurements, the sample was exposed to an external force. Once the desired constant strain was achieved, the stress quickly reduced until it reached a stable level, which reflected the viscoelastic behavior of the sample (Ozturk and Singh Takhar, 2019; Yang et al., 2022). Overall, the stress relaxation curves of Chinese yams exhibited similar trends under various microwave power levels. The Maxwell model was utilized to describe the stress relaxation behavior, with fitting parameters (E_0 , E_1 , τ , and n) and the coefficient of determination (R^2) of the model listed in Table 1. The experimental data exhibited a well fit with the model, as evidenced by $R^2 > 0.99$ for each condition. These results suggested that the Maxwell model effectively described the stress relaxation behavior of Chinese yam.

Throughout the drying period, the relaxation modulus of yam showed a trend of an initial decrease, later followed by a rise, which was closely linked to its internal structural evolution. Fresh yam, characterized by a dense internal structure and high moisture content, exhibited a higher relaxation modulus. During the early drying stage,

Table 1
Maxwell model parameters.

	Drying time (min)	E_0 (MPa)	E_1 (MPa)	n	τ	R^2
2 W/g	0	87.289 ± 2.825 ^b	57.140 ± 13.736 ^b	0.371 ± 0.008 ^a	3.701 ± 0.060 ^a	0.999
	8	86.846 ± 2.651 ^b	52.820 ± 4.744 ^{bc}	0.358 ± 0.006 ^a	3.076 ± 0.278 ^b	0.999
	16	22.112 ± 1.389 ^e	22.982 ± 3.865 ^d	0.342 ± 0.032 ^{ab}	2.053 ± 0.261 ^c	0.998
	24	19.998 ± 1.326 ^{ef}	22.525 ± 1.621 ^d	0.317 ± 0.018 ^{bc}	2.045 ± 0.397 ^c	0.998
	32	13.481 ± 1.969 ^f	18.051 ± 2.290 ^d	0.309 ± 0.001 ^{bc}	2.077 ± 0.043 ^c	0.998
	40	30.006 ± 0.743 ^d	35.096 ± 2.382 ^{cd}	0.305 ± 0.003 ^c	2.136 ± 0.146 ^c	0.997
	48	39.015 ± 3.581 ^c	54.104 ± 1.008 ^b	0.303 ± 0.008 ^c	2.075 ± 0.098 ^c	0.997
	56	97.826 ± 7.567 ^a	78.333 ± 15.976 ^a	0.321 ± 0.004 ^{bc}	2.414 ± 0.042 ^c	0.999
4 W/g	0	87.289 ± 2.825 ^b	57.140 ± 13.736 ^b	0.371 ± 0.008 ^a	3.701 ± 0.060 ^a	0.999
	4	34.848 ± 1.769 ^c	57.128 ± 0.823 ^b	0.324 ± 0.016 ^b	1.861 ± 0.117 ^c	0.996
	8	6.928 ± 0.614 ^e	10.459 ± 1.514 ^d	0.299 ± 0.019 ^{bc}	1.627 ± 0.091 ^c	0.996
	12	16.901 ± 0.930 ^d	36.334 ± 8.682 ^c	0.270 ± 0.018 ^{cd}	1.624 ± 0.221 ^c	0.996
	16	36.528 ± 1.636 ^c	78.739 ± 0.105 ^a	0.259 ± 0.001 ^d	1.595 ± 0.030 ^c	0.994
	20	122.442 ± 2.325 ^a	92.905 ± 1.599 ^a	0.320 ± 0.010 ^b	2.618 ± 0.158 ^b	0.999
	0	87.289 ± 2.825 ^{bc}	57.140 ± 13.736 ^c	0.371 ± 0.008 ^a	3.701 ± 0.060 ^a	0.999
	2	63.653 ± 0.148 ^{cd}	55.462 ± 0.023 ^c	0.337 ± 0.011 ^{ab}	3.504 ± 0.358 ^a	0.999
6 W/g	4	28.926 ± 2.297 ^{de}	29.627 ± 4.343 ^{de}	0.332 ± 0.005 ^{ab}	2.134 ± 0.088 ^b	0.998
	6	22.016 ± 3.811 ^e	23.275 ± 1.472 ^e	0.328 ± 0.008 ^{ab}	2.325 ± 0.182 ^b	0.999
	8	30.220 ± 1.466 ^{de}	27.841 ± 0.022 ^{de}	0.315 ± 0.001 ^{abc}	2.225 ± 0.037 ^b	0.998
	10	69.548 ± 5.706 ^c	47.932 ± 0.371 ^{cd}	0.310 ± 0.001 ^{bc}	2.608 ± 0.046 ^b	0.999
	12	119.335 ± 2.205 ^b	92.801 ± 12.304 ^b	0.271 ± 0.023 ^{cd}	2.456 ± 0.254 ^b	0.999
	14	186.241 ± 41.602 ^a	128.558 ± 18.489 ^a	0.254 ± 0.058 ^d	2.328 ± 0.616 ^b	0.998
	0	87.289 ± 2.825 ^d	57.140 ± 13.736 ^c	0.371 ± 0.008 ^a	3.701 ± 0.060 ^a	0.999
	2	77.493 ± 6.730 ^d	91.555 ± 2.552 ^c	0.328 ± 0.002 ^a	2.217 ± 0.047 ^c	0.998
8 W/g	4	67.322 ± 6.157 ^d	78.787 ± 12.109 ^c	0.313 ± 0.001 ^a	2.178 ± 0.247 ^c	0.997
	6	166.499 ± 7.543 ^c	147.470 ± 21.361 ^b	0.306 ± 0.006 ^a	2.770 ± 0.281 ^b	0.998
	8	252.462 ± 24.228 ^b	177.364 ± 26.809 ^b	0.321 ± 0.068 ^a	3.031 ± 0.205 ^b	0.999
	10	324.826 ± 46.860 ^a	345.865 ± 18.816 ^a	0.322 ± 0.010 ^a	3.069 ± 0.238 ^b	0.999
	0	87.289 ± 2.825 ^c	57.140 ± 13.736 ^c	0.371 ± 0.008 ^a	3.701 ± 0.060 ^a	0.999
	2	34.664 ± 5.117 ^d	23.422 ± 2.716 ^d	0.359 ± 0.001 ^a	3.091 ± 0.027 ^b	0.999
	4	22.425 ± 0.045 ^d	23.156 ± 0.378 ^d	0.348 ± 0.003 ^{ab}	2.794 ± 0.156 ^b	0.998
	6	282.615 ± 39.641 ^b	275.069 ± 15.652 ^b	0.340 ± 0.004 ^{ab}	3.027 ± 0.116 ^b	0.999
10 W/g	8	535.189 ± 0.834 ^a	389.671 ± 7.588 ^a	0.304 ± 0.041 ^b	2.969 ± 0.454 ^b	0.999

Different letters represent significant difference between the values at different drying time ($P < 0.05$).

Table 2

Correlation analysis of shrinkage with NMR parameters.

		A_{21}	A_{22}	A_{23}	A
2 W/g	VSR	-0.530*	0.073	-0.981**	-0.973**
	TPASR	-0.348	0.255	-0.985**	-0.898**
4 W/g	VSR	-0.852**	0.137	-0.960**	-0.984**
	TPASR	-0.778**	0.263	-0.981**	-0.949**
6 W/g	VSR	-0.869**	-0.208	-0.903**	-0.980**
	TPASR	-0.871**	-0.058	-0.958**	-0.952**
8 W/g	VSR	-0.939**	-0.121	-0.977**	-0.922**
	TPASR	-0.948**	-0.196	-0.982**	-0.958**
10 W/g	VSR	-0.487	-0.022	-0.988**	-0.907**
	TPASR	-0.442	-0.168	-0.961**	-0.947**

** $P < 0.01$ Correlation is significant at the 0.01 level; * $P < 0.05$ Correlation is significant at the 0.05 level.

the plasticizing influence of liquid water enhanced the mobility of molecular chain, leading to sample softening. Consequently, this caused a significant reduction in relaxation modulus. This observation aligned with the changes in textural properties observed in the dried samples. De Bruijn and Bórquez (2014) also reported that such changes in dried materials, including cell wall modifications, tissue damage, and loss of hydrostatic pressure, contributed to reduced mechanical properties, which diminished the resistance to internal shrinkage stresses. As a result, the samples became prone to deformation. As drying progressed, the formation of internal pore and skeletal structures within the sample enhanced its rigidity. Simultaneously, the significant reduction in MC raised the glass transition temperature of the sample, triggering a phase transition from a rubbery state to a glassy state. This phase transition further contributed to the increase in rigidity. The combined effects of these factors caused an increase in the relaxation modulus, enabling the sample to effectively resist drying-induced stresses and thereby inhibiting deformation.

At the end of drying, the E_0 and E_1 of the samples treated at 10 W/g increased by 437.363 and 311.338, respectively, compared to those treated at 2 W/g, while the VSR significantly decreased by 9.011 % ($P < 0.05$). This phenomenon indicated that under high microwave power levels, the samples developed a more rigid structure, which effectively resisted externally applied mechanical forces, thereby inhibiting shrinkage deformation. The relaxation index (n) is correlated with the structural stability of the sample network and its resistance to external forces (Li et al., 2022b). A noticeable reduction in n values was observed in MPD treated samples, indicating that the internal structural integrity of the samples was partially damaged during microwave heating, leading to reduced resistance to external stresses. The relaxation time (τ), which comprehensively reflects the viscoelastic behavior of materials (Yang et al., 2025), exhibited reductions ranging from 17.07 % to 37.09 % at the end of drying when compared to fresh samples (Table 1). This phenomenon could be ascribed to the evolving moisture states and microporous structure during thermal processing, which dynamically altered viscoelastic characteristics. Concurrently, water removal enhanced the solid matrix's deformation resistance. Through systematic investigation of relaxation modulus variations under different MPD conditions, it revealed that higher microwave power not only accelerated drying kinetics but also improved shrinkage deformation of final products through rigidity enhancement.

3.3.3. Moisture migration behavior analysis

Fig. 8(a) shows the variations in T_2 curves of yam slices during MVD under different MPDs. Three distinct relaxation peaks were identified in fresh yam: T_{21} (0.1–10 ms), representing tightly bound water associated with macromolecular substances such as cell wall polysaccharides; T_{22} (10–100 ms), corresponding to immobilized water trapped within highly organized structures; and T_{23} (100–1000 ms), attributed to highly mobile free water residing in vacuoles or extracellular spaces. T_2 reflects the water-binding capacity and the level of freedom of hydrogen protons

in the sample. A shorter T_2 indicates lower freedom of hydrogen protons and stronger water-binding interactions. For different microwave power levels, the T_2 curves demonstrated a trend of shifting toward lower relaxation times during MVD, indicating a gradual reduction in water mobility as drying progressed.

Fig. 8(b) displays the variations in the signal intensities of relaxation peak during MVD under different MPDs. The signal intensity of relaxation peaks, quantified by its peak area, reflects the content of water in distinct states. Here, A_{21} , A_{22} , and A_{23} correspond to the areas of T_{21} , T_{22} , and T_{23} relaxation peaks, representing the contents of bound water, immobilized water, and free water, respectively. The symbol A denotes the sum of A_{21} , A_{22} , and A_{23} . In fresh yam, free water constituted the dominant component, with A_{23} accounting for approximately 74 % of A . Consequently, the evolution of A was primarily governed by changes in A_{23} . Under all MPD conditions, the value of A exhibited a continuous decline with prolonged drying time, which was in line with the gradual loss of moisture during the drying process. Notably, samples dried at higher MPD levels demonstrated faster moisture loss compared to those processed at lower MPD levels. This difference can be attributed to the rapid and intensive heating effects at higher microwave power.

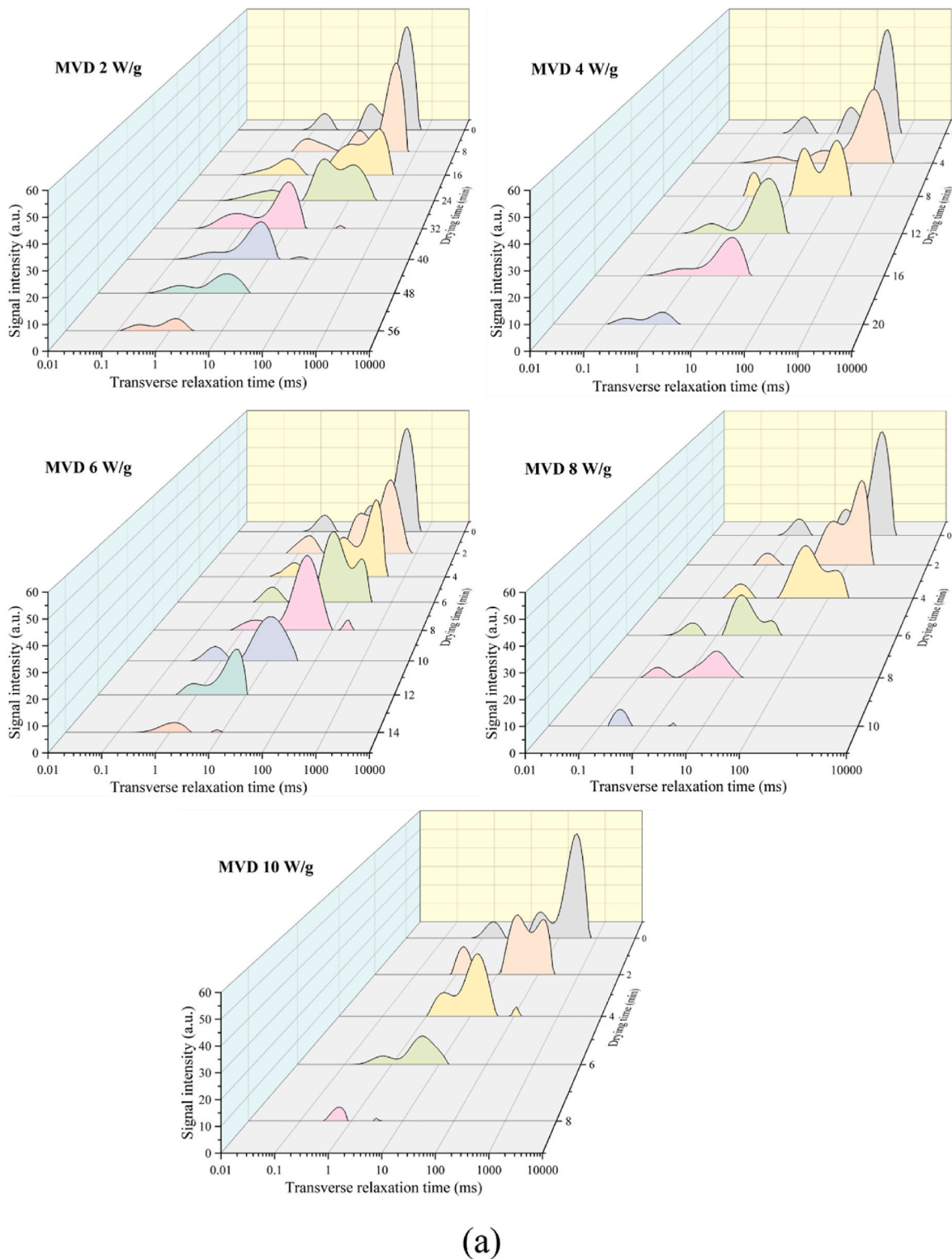
A_{23} exhibited a similar variation pattern to A , indicating that free water was more mobile and was preferentially removed during the drying process. In the initial drying stage, the continuous outward migration of free water from vacuoles and extracellular spaces led to a macroscopic dimensional shrinkage (Liu et al., 2022). This phenomenon can be due to the loss of free water, which eliminates structural support for cell walls and causes the collapse of intercellular pore networks, thereby inducing macroscopic shrinkage. This study by Sun et al. (2022) further supported this mechanism, demonstrating that drying-induced shrinkage primarily originated from moisture loss in vacuolar compartments. Table 2 presents the correlations between shrinkage ratios and NMR parameters (A_{21} , A_{22} , A_{23} , and A) under varying MPDs. Pearson correlation analysis revealed statistically significant negative relationships ($r > 0.90$, $P < 0.01$) between dimensional changes and both A_{23} and A , validating that the loss of free water and total water were the critical determinants of shrinkage behavior in Chinese yam samples.

The A_{22} value exhibited a generally increasing trend prior to the shrinkage inflection point. It resulted from the progressive removal of free water, which increased the intracellular solute concentration, causing some of the free water to be trapped within the sample tissue matrix and transformed into less mobile immobilized water. Following the shrinkage inflection point, the immobilized water content gradually decreased, becoming the primary factor influencing volume shrinkage. On one hand, this water was converted into free water and subsequently removed; on the other hand, drying-induced cell membrane damage likely facilitated the transformation of immobilized water into water with even lower mobility through interactions with macromolecules (Li et al., 2018), further contributing to sample shrinkage. Khan and Karim (2017) also noted that the migration of immobilized water led to cell contraction, pore formation, and cellular collapse.

3.4. Physical and chemical properties analysis

3.4.1. Color

Color is a critical quality attribute of dried products, directly affecting consumer perception and purchase preference. Fig. 4(c) presents the color parameter variations of Chinese yam dried under different MPDs. Overall, with increasing MPD, the L^* and a^* values showed an upward trend, while the b^* value decreased, indicating a transition in color from dark yellow to bright white. At the low power density of 2 W/g, the samples exhibited significantly lower L^* and a^* values and a significantly higher b^* value compared to other power levels ($P < 0.05$), suggesting a darker and more yellowish appearance. This was attributed to the prolonged drying time under low power, which intensified the Maillard reaction and led to the accumulation of

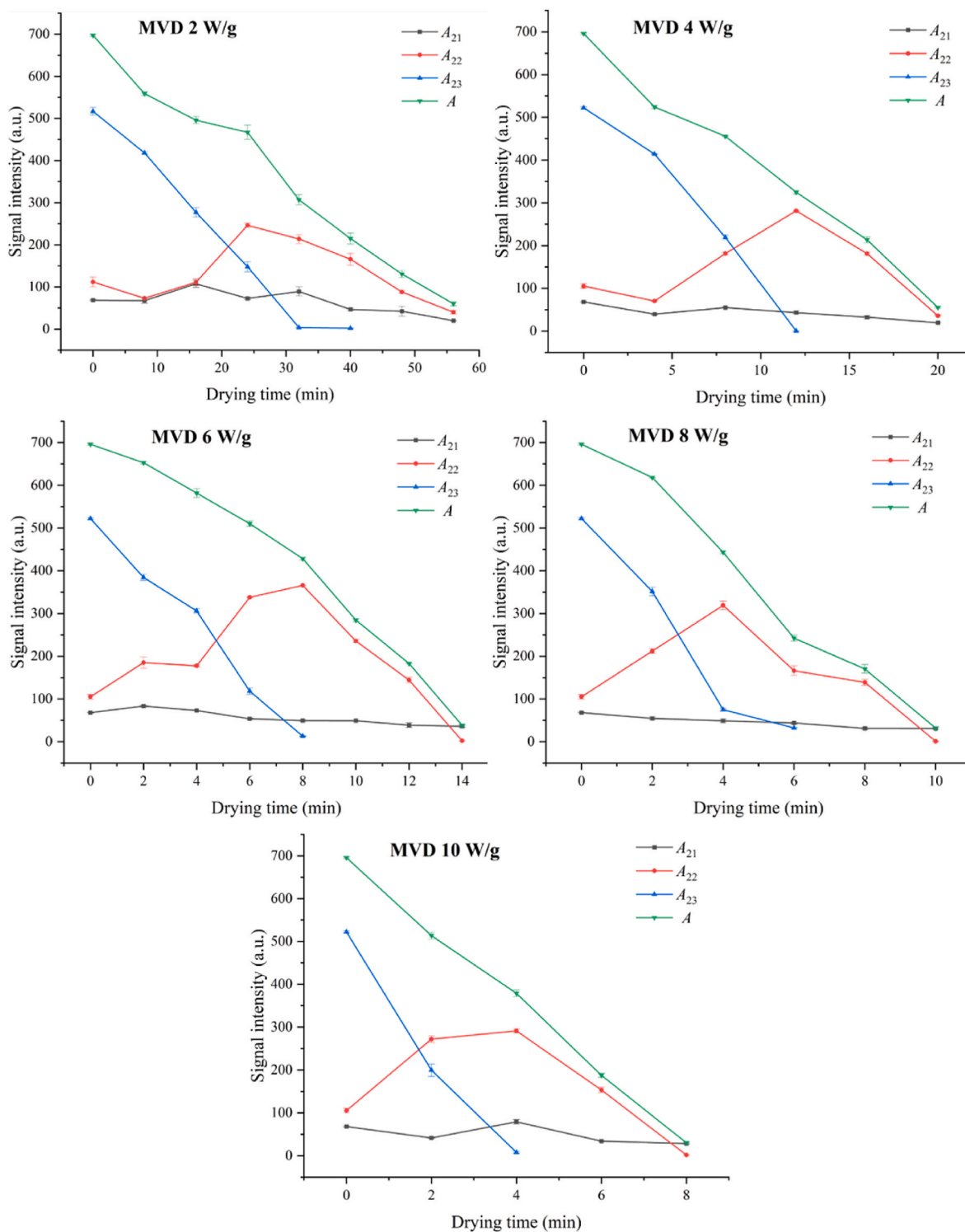


(a)

Fig. 8. Moisture migration behavior of yams under different drying conditions: (a) transverse relaxation time (T_2) curves, (b) NMR parameters of yam during MVD.

yellow-brown pigments, resulting in a darker yellow appearance (Xu et al., 2022a). In contrast, the sample dried at 10 W/g exhibited the most desirable color characteristics, with the highest L^* value of 88.875, an a^* value of 0.867, and the lowest b^* value of 8.558, indicating a brighter, whiter color with reduced yellowness. Moreover, no significant differences were observed in the color parameters of samples dried at higher

power densities (6, 8, and 10 W/g), suggesting that these conditions better preserved the original color attributes of Chinese yam. This could be attributed to the rapid drying process under high microwave power, which effectively shortened the exposure time to high temperatures and thereby inhibited enzymatic and non-enzymatic browning reactions, reducing color degradation (Zhang et al., 2024). Therefore, from the



(b)

Fig. 8. (continued).

perspective of color quality, high MPD was more favorable for maintaining the desirable appearance of dried Chinese yam.

3.4.2. TPC and TFC

Fig. 4(d) illustrates the variations in TPC and TFC of Chinese yam dried under different MPD. As the MPD increased, both TPC and

TFC exhibited a decreasing-then-increasing trend. The TPC and TFC values across different power levels ranged from 52.41 to 86.80 mg GAE/100 g DW and 100.43–128.69 mg RE/100 g DW, respectively. Among them, the sample dried at 2 W/g showed significantly higher TPC and TFC values than those dried at other power levels ($P < 0.05$), with TPC and TFC reaching 86.80 mg GAE/100 g DW and 128.69 mg RE/100 g DW,

respectively. This indicated that low MPD was conducive to the retention of phenolic and flavonoid compounds, possibly because the mild heating conditions reduced thermal degradation and oxidation. In addition, microwave irradiation disrupted covalent bonds in intracellular polymers during drying, thereby enhancing the release and extractability of these bioactive compounds (Xu et al., 2022b). With the further increase in MPD, the rapid temperature rise might cause partial degradation of active compounds due to structural damage. However, microwave heating under vacuum conditions led to accelerated and intensified heating, which not only reduced the activity of polyphenol oxidase but also significantly shortened the drying time, thus preventing excessive degradation of bioactive compounds. Consequently, the retention of TPC and TFC increased at higher power levels (Xu et al., 2022a; Zhang et al., 2024).

Based on the above findings, although the highest levels of bioactive compounds were observed at 2 W/g, this condition was associated with longer drying time, inferior color quality, and greater shrinkage. Therefore, from the perspective of industrial application, a power density of 8 W/g is more suitable for efficient production of high-quality dried Chinese yam.

4. Conclusions

In this study, the shrinkage deformation behavior of Chinese yam during MVD was investigated. Through comprehensive analyses of textural properties, stress relaxation behavior, and moisture migration patterns, the intrinsic mechanisms underlying the shrinkage phenomena in Chinese yam were elucidated. The results demonstrated that shrinkage induced by drying resulted from the combined effects of radial and longitudinal shrinkage. The shrinkage behavior, evaluated by TPASR and VSR, of Chinese yam during MVD exhibited a two-stage pattern: an initial rapid shrinkage phase followed by a slower shrinkage phase, with a distinct inflection point related to moisture content. The mechanism can be explained as follows: in the early drying stage, samples maintained a rubber state and has high moisture content, which exhibited insufficient mechanical strength to resist the stress induced by moisture and temperature gradients, thereby leading to rapid shrinkage. As drying progressed, the gradual increase in hardness and brittleness enhanced the material's resistance to drying stress, thereby reducing subsequent shrinkage. This mechanism was strongly supported by the observed trends in textural and viscoelastic properties, both of which demonstrated an initial decrease followed by a subsequent increase. Furthermore, a significant linear correlation ($R^2 > 0.95$, $P < 0.01$) was identified between TPASR and moisture content throughout the drying process.

Studies on the effect of microwave power levels showed that the impacts of MPD levels on shrinkage were predominantly in longitudinal contraction. The volumetric shrinkage rate decreased with increasing MPD. Samples dried at low MPDs underwent severe deformation, characterized by central collapse of contour edges and peripheral curling, whereas high MPDs effectively suppressed longitudinal shrinkage. In addition, samples dried at high microwave power levels exhibited favorable color characteristics and retained a certain level of bioactive compounds. It should be noted that although this study determined the optimal microwave power density to be 8 W/g, during the later stages of drying, to prevent overheating, the process was still subject to the automatic control system of the testing equipment, which may have had a certain impact on the shrinkage behavior observed in the final stage.

However, the mechanisms of moisture migration on microstructural evolution in Chinese yam during MVD are still unclear, and the mechanism of the resulting mediated changes in mechanical properties on the shrinkage behavior in plant-based materials requires further investigation in future research.

CRediT authorship contribution statement

Qian Zhou: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Linlin Li:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lijuan Wang:** Supervision, Software, Methodology, Investigation. **Wenchao Liu:** Supervision, Methodology, Funding acquisition, Formal analysis. **Weiwei Cao:** Supervision, Methodology, Funding acquisition. **Tongxiang Yang:** Supervision, Methodology. **Junliang Chen:** Methodology, Funding acquisition. **Guangyue Ren:** Supervision, Funding acquisition. **Chung Lim Law:** Supervision, Methodology. **Xu Duan:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this article.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 32302285, No. 32172352), Leading Talent Program for Science and Technology Innovation in Central China (No. 234200510020), Natural Science Foundation of Henan (No. 242300421563), Luoyang Scientific and Technological Research Project (No. 2302021A), all of which enabled us to carry out this study.

Data availability

Data will be made available on request.

References

- Ando, Y., Hagiwara, S., Nabetani, H., Sotome, I., Okunishi, T., Okadome, H., Orikasa, T., Tagawa, A., 2019. Effects of prefreezing on the drying characteristics, structural formation and mechanical properties of microwave-vacuum dried apple. *J. Food Eng.* 244, 170–177. <https://doi.org/10.1016/j.jfoodeng.2018.09.026>.
- Ando, Y., Nei, D., 2024. Effects of different pre-freezing methods and pressure levels on the pore structure and mechanical properties of microwave-vacuum dried apple. *J. Food Eng.* 369, 111944. <https://doi.org/10.1016/j.jfoodeng.2024.111944>.
- Aprajeta, J., Gopirajah, R., Anandharamakrishnan, C., 2015. Shrinkage and porosity effects on heat and mass transfer during potato drying. *J. Food Eng.* 144, 119–128. <https://doi.org/10.1016/j.jfoodeng.2014.08.004>.
- Chao, E., Li, J., Fan, L., 2023. Influence of combined freeze-drying and far-infrared drying technologies on physicochemical properties of seed-used pumpkin. *Food Chem.* 398, 133849. <https://doi.org/10.1016/j.foodchem.2022.133849>.
- De Bruijn, J., Bórquez, R., 2014. Quality retention in strawberries dried by emerging dehydration methods. *Food Res. Int.* 63, 42–48. <https://doi.org/10.1016/j.foodres.2014.03.029>.
- Duan, L., Duan, X., Ren, G., 2020. Structural characteristics and texture during the microwave freeze drying process of Chinese yam chips. *Dry. Technol.* 38 (7), 928–939. <https://doi.org/10.1080/07373937.2019.1600142>.
- Duan, X., Yang, X., Ren, G., Pang, Y., Liu, L., Liu, Y., 2016. Technical aspects in freeze-drying of foods. *Dry. Technol.* 34 (11), 1271–1285. <https://doi.org/10.1080/07373937.2015.1099545>.
- Gulati, T., Datta, A.K., 2015. Mechanistic understanding of case-hardening and texture development during drying of food materials. *J. Food Eng.* 166, 119–138. <https://doi.org/10.1016/j.jfoodeng.2015.05.031>.
- He, X., Lin, R., Cheng, S., Wang, S., Yuan, L., Wang, H., Wang, H., Tan, M., 2021. Effects of microwave vacuum drying on the moisture migration, microstructure, and rehydration of sea cucumber. *J. Food Sci.* 86 (6), 2499–2512. <https://doi.org/10.1111/1750-3841.15716>.
- Huang, L.L., Lian, M.M., Duan, X., Li, B., Yang, S.Z., 2021. Studies on the quality and moisture distribution of kiwifruit dried by freeze drying combined with microwave vacuum drying. *J. Food Process. Eng.* 44 (1), e13581. <https://doi.org/10.1111/jfpe.13581>.
- Ishibashi, R., Numata, T., Tanigawa, H., Tsuruta, T., 2022. In-situ measurements of drying and shrinkage characteristics during microwave vacuum drying of radish and potato. *J. Food Eng.* 323, 110988. <https://doi.org/10.1016/j.jfoodeng.2022.110988>.

- Khan, M.I.H., Karim, M.A., 2017. Cellular water distribution, transport, and its investigation methods for plant-based food material. *Food Res. Int.* 99, 1–14. <https://doi.org/10.1016/j.foodres.2017.06.037>.
- Kurozawa, L.E., Hubinger, M.D., Park, K.J., 2012. Glass transition phenomenon on shrinkage of papaya during convective drying. *J. Food Eng.* 108 (1), 43–50. <https://doi.org/10.1016/j.jfoodeng.2011.07.033>.
- Li, G., Li, L., Liu, W., Cao, W., Chen, J., Ren, G., Law, C.L., Duan, X., 2024. Effects of multiphase microwave drying on the microstructure of Chinese yam: a study based on X-ray micro-computed tomography technology. *LWT* 203, 116327. <https://doi.org/10.1016/j.lwt.2024.116327>.
- Li, L., Chen, J., Bai, D., Xu, M., Cao, W., Ren, G., Ren, A., Duan, X., 2022a. Physicochemical, pasting properties and in vitro starch digestion of Chinese yam flours as affected by microwave freeze-drying. *Foods* 11 (15), 2324. <https://doi.org/10.3390/foods11152324>.
- Li, L., Ren, X., Chen, J., Cao, W., Ren, G., Bhandari, B., Ren, A., Duan, X., 2022b. Changes and relationships of viscoelastic and physical properties of Chinese yam during a novel multiphase microwave drying process. *LWT* 168, 113969. <https://doi.org/10.1016/j.lwt.2022.113969>.
- Li, L., Zhang, M., Bhandari, B., Zhou, L., 2018. LF-NMR online detection of water dynamics in apple cubes during microwave vacuum drying. *Dry. Technol.* 36 (16), 2006–2015. <https://doi.org/10.1080/07373937.2018.1432643>.
- Li, L., Zhang, M., Yang, P., 2019. Suitability of LF-NMR to analysis water state and predict dielectric properties of Chinese yam during microwave vacuum drying. *LWT* 105, 257–264. <https://doi.org/10.1016/j.lwt.2019.02.017>.
- Li, L., Zhang, M., Zhou, L., 2021. A promising pulse-spouted microwave freeze drying method used for Chinese yam cubes dehydration: quality, energy consumption, and uniformity. *Dry. Technol.* 39 (2), 148–161. <https://doi.org/10.1080/07373937.2019.1624564>.
- Li, Y., Ji, S., Xu, T., Zhong, Y., Xu, M., Liu, Y., Li, M., Fan, B., Wang, F., Xiao, J., Lu, B., 2023. Chinese yam (*Dioscorea*): nutritional value, beneficial effects, and food and pharmaceutical applications. *Trends Food Sci. Technol.* 134, 29–40. <https://doi.org/10.1016/j.tifs.2023.01.021>.
- Liu, Y.-y., Sun, W.-h., Li, B.-z., Shang, N., Wang, Y., Lv, W.-q., Li, D., Wang, L.-j., 2021. Value-added application of *Platycodon grandiflorus* (Jacq.) A.DC. roots (PGR) by ultrasound-assisted extraction (UAE) process to improve physicochemical quality, structural characteristics and functional properties. *Food Chem.* 363, 130354. <https://doi.org/10.1016/j.foodchem.2021.130354>.
- Liu, Y.-y., Sun, W.-h., Li, B.-z., Wang, Y., Lv, W.-q., Shang, N., Li, D., Wang, L.-j., 2022. Dehydration characteristics and evaluation of physicochemical properties of *Platycodon grandiflorum* (Jacq. A.DC.) roots (PGR) during pulse-spouted microwave vacuum drying (PSMVD). *Ind. Crop. Prod.* 177, 114449. <https://doi.org/10.1016/j.indcrop.2021.114449>.
- Mahiuddin, M., Khan, M.I.H., Kumar, C., Rahman, M.M., Karim, M.A., 2018. Shrinkage of food materials during drying: current status and challenges. *Compr. Rev. Food Sci. Food Saf.* 17 (5), 1113–1126. <https://doi.org/10.1111/1541-4337.12375>.
- Mayor, L., Sereno, A.M., 2004. Modelling shrinkage during convective drying of food materials: a review. *J. Food Eng.* 61 (3), 373–386. [https://doi.org/10.1016/S0260-8774\(03\)00144-4](https://doi.org/10.1016/S0260-8774(03)00144-4).
- Monteiro, R.L., Link, J.V., Tribuzzi, G., Carciofi, B.A.M., Laurindo, J.B., 2018. Effect of multi-flash drying and microwave vacuum drying on the microstructure and texture of pumpkin slices. *LWT* 96, 612–619. <https://doi.org/10.1016/j.lwt.2018.06.023>.
- Nguyen, T.K., Khalloufi, S., Mondor, M., Ratti, C., 2018. Shrinkage and porosity evolution during air-drying of non-cellular food systems: experimental data versus mathematical modelling. *Food Res. Int.* 103, 215–225. <https://doi.org/10.1016/j.foodres.2017.10.013>.
- Ortiz-García-Carrasco, B., Yañez-Mota, E., Pacheco-Aguirre, F.M., Ruiz-Espinosa, H., García-Alvarado, M.A., Cortés-Zavaleta, O., Ruiz-López, I.I., 2015. Drying of shrinkable food products: appraisal of deformation behavior and moisture diffusivity estimation under isotropic shrinkage. *J. Food Eng.* 144, 138–147. <https://doi.org/10.1016/j.jfoodeng.2014.07.022>.
- Ozturk, O.K., Singh Takhar, P., 2019. Selected physical and viscoelastic properties of strawberries as a function of heated-air drying conditions. *Dry. Technol.* 37 (14), 1833–1843. <https://doi.org/10.1080/07373937.2018.1543701>.
- Ren, X., Li, L., Chen, J., Zhao, L., Liu, P., Cao, W., Ren, A., Ren, G., Bhandari, B., Duan, X., 2022. Drying characteristics and quality of Chinese yam by multiphase microwave drying based on fractal theory. *Dry. Technol.* 40 (15), 3310–3323. <https://doi.org/10.1080/07373937.2021.2024220>.
- Sappati, P.K., Nayak, B., Van Walsum, G.P., 2017. Effect of glass transition on the shrinkage of sugar kelp (*Saccharina latissima*) during hot air convective drying. *J. Food Eng.* 210, 50–61. <https://doi.org/10.1016/j.jfoodeng.2017.04.018>.
- Seerangurayar, T., Al-Ismailli, A.M., Janitha Jeewantha, L.H., Al-Nabhani, A., 2019. Experimental investigation of shrinkage and microstructural properties of date fruits at three solar drying methods. *Sol. Energy* 180, 445–455. <https://doi.org/10.1016/j.solener.2019.01.047>.
- Sun, Q., Zhang, M., Mujumdar, A.S., Yu, D., 2022. Research on the vegetable shrinkage during drying and characterization and control based on LF-NMR. *Food Bioprocess Technol.* 15 (12), 2776–2788. <https://doi.org/10.1007/s11947-022-02917-x>.
- Sun, Y., Zhang, M., Mujumdar, A.S., Yu, D., 2021. Pulse-spouted microwave freeze drying of raspberry: control of moisture using ANN model aided by LF-NMR. *J. Food Eng.* 292, 110354. <https://doi.org/10.1016/j.jfoodeng.2020.110354>.
- Vallespir, F., Crescenzo, L., Rodríguez, Ó., Marra, F., Simal, S., 2019. Intensification of low-temperature drying of mushroom by means of power ultrasound: effects on drying kinetics and quality parameters. *Food Bioprocess Technol.* 12 (5), 839–851. <https://doi.org/10.1007/s11947-019-02263-5>.
- Wang, H., Liu, C., Xue, Y., Li, D., 2020. Correlation of mechanical properties of peach slices with cell wall polysaccharides and cell morphology during hot air predrying. *J. Food Process. Preserv.* 44 (2). <https://doi.org/10.1111/jfpp.14319>.
- Xu, B., Feng, M., Chitrakar, B., Wei, B., Wang, B., Zhou, C., Ma, H., Wang, B., Chang, L., Ren, G., Duan, X., 2022a. Selection of drying techniques for Pingyin rose on the basis of physicochemical properties and volatile compounds retention. *Food Chem.* 385, 132539. <https://doi.org/10.1016/j.foodchem.2022.132539>.
- Xu, Q., Pan, H., Shui, Y., Xing, Y., Wu, L., Zheng, F., Fan, X., Bi, X., 2022b. Effect of different drying technologies on the characteristics and quality of lemon slices. *J. Food Sci.* 87 (7), 2980–2998. <https://doi.org/10.1111/1750-3841.16194>.
- Yang, M., Chen, X., Wang, L., Ren, X., Liu, W., Li, L., Ren, G., Law, C.L., Cao, W., Zhang, M., Chen, J., Lu, D., Duan, X., 2025. Effect of moisture migration on texture and viscoelasticity of Chinese yam during multi-phase microwave drying (MMD) - focusing on the evaporation-dehydration stage. *Food Chem. X* 26, 102314. <https://doi.org/10.1016/j.fochx.2025.102314>.
- Yang, M., Li, L., Liu, W., Cao, W., Chen, J., Ren, G., Gao, K., Law, C.L., Duan, X., 2024. An evaluation on the shrinkage deformation characteristics of Chinese yam during multiphase microwave drying based on digital image processing. *Innov. Food Sci. Emerg. Technol.* 95, 103725. <https://doi.org/10.1016/j.ifset.2024.103725>.
- Yang, Y.-L., Guan, E.-q., Zhang, L.-L., Li, M.-m., Bian, K., 2022. Mechanical action on the development of dough and its influence on rheological properties and protein network structure. *Food Res. Int.* 158, 111495. <https://doi.org/10.1016/j.foodres.2022.111495>.
- Yuan, Y., Tan, L., Xu, Y., Yuan, Y., Dong, J., 2019. Numerical and experimental study on drying shrinkage-deformation of apple slices during process of heat-mass transfer. *Int. J. Therm. Sci.* 136, 539–548. <https://doi.org/10.1016/j.jthermalsci.2018.10.042>.
- Zhang, H., Zheng, W., Yan, L., Liu, W., Yao, F., Liu, C., Zheng, L., 2023. Effects of vacuum microwave combined with freeze-drying on the physicochemical properties, phenolic compounds, and antioxidant capacity of pear fruit slices. *J. Food Sci.* 88 (7), 2807–2820. <https://doi.org/10.1111/1750-3841.16653>.
- Zhang, M., Chen, H., Mujumdar, A.S., Tang, J., Miao, S., Wang, Y., 2017. Recent developments in high-quality drying of vegetables, fruits, and aquatic products. *Crit. Rev. Food Sci. Nutr.* 57 (6), 1239–1255. <https://doi.org/10.1080/10408398.2014.979280>.
- Zhang, M., Wu, C., Zhang, H., Yang, N., Wang, C., Jike, X., Zhang, T., Lei, H., 2024. Comparison of different drying technologies for kiwifruit pomace: changes in physical characteristics, nutritional properties and antioxidant capacities. *Food Chem.* 451, 139497. <https://doi.org/10.1016/j.foodchem.2024.139497>.
- Zielinska, M., Zielinska, D., Markowski, M., 2018. The effect of microwave-vacuum pretreatment on the drying kinetics, color and the content of bioactive compounds in osmo-microwave-vacuum dried Cranberries (*Vaccinium macrocarpon*). *Food Bioprocess Technol.* 11 (3), 585–602. <https://doi.org/10.1007/s11947-017-2034-9>.